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Dark personality traits are associated with academic misconduct, frustration, negative thinking, and generative AI use habits: the case of Sichuan art universities

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Abstract

Background Academic misconduct among students is common, particularly now that AI tools are widely available. Students with dark personality traits may be more likely to engage in misconduct. Drawing on social cognitive theory, this study examines how the dark personality traits of materialism, narcissism, psychopathy, and Machiavellianism are associated with academic dishonesty, anxiety, and procrastination, and, in turn, frustration, negative thinking and generative AI use habits.

Method Self-reported data from 504 Chinese university art students were analyzed using structural equation modeling.

Results Results showed that dark personality traits were significantly associated with academic dishonesty, anxiety, and procrastination, and that these markers of academic misbehavior were linked to increased frustration, negative thinking, and generative AI use.

Conclusion The findings highlight how personality traits are related to problematic academic behaviors and psychological difficulties. Academic institutions should provide appropriate training and psychological support to counter academic dishonesty and ensure students understand the value of academic ethics.

Keywords Dark personality traits, Academic misconduct, Dishonesty, Procrastination, Generative AI use

Introduction

Academic integrity is a cornerstone of higher education, yet ensuring integrity is becoming more difficult due to technological advances, particularly in generative AI tools such as ChatGPT, DALL-E, Midjourney, and Stable Diffusion [1]. These tools offer students access to

resources that can make academic tasks easier but also give rise to plagiarism, cheating, and unethical use of AI in learning environments [2]. As AI tools make it easier for students to cheat or bypass traditional approaches to learning and creation, universities around the world are struggling to maintain academic standards and integrity [3]. While AI in education is not new, its widespread recent adoption has introduced new paths to academic dishonesty while increasing existing pressures on students [4].

Academic misconduct is defined as any act that violates the academic integrity and ethical standards of

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educational activities [5]. Art students in China face specific academic problems that may elevate the likelihood they engage in misconduct. Intense competition is part of China's education system, especially in fields such as art and design where students compete for a small number of places at prestigious institutions [6]. China's system of dividing students into academic and vocational tracks adds another level of complexity. Art students are typically placed in vocational tracks and put under pressure to show both creative output and technical knowledge to bolster their academic worth [7]. Facing pressures of competition and high expectations for originality and innovation, students often become stressed and anxious, and may resort to dishonesty when trying to meet artistic and academic demands [5].

It is important to comprehend the academic behaviors of art students in this regard because these learners face the problem of mastering conventional artistic skills while at the same time incorporating new technologies into their practice. Also, the nature of art education, whereby students are judged based on perceptions of their work, can lead to significant stress if they do not perform as expected. Such circumstances may lead to unethical practices such as plagiarism or using generative AI for assignments. These pressures might be amplified by dark triad personality traits such as narcissism and Machiavellianism, which are associated with academic dishonesty [8].

Generative AI tools such as DALL-E and Midjourney are changing art and design education while raising new issues around cheating. These technologies enable learners to produce extremely complex artwork with virtually little effort, thereby reducing the distinction between creativity and artificial intelligence [1]. For instance, a student can type commands to an AI tool to generate an artistic and appealing picture within seconds, no longer needing to draw or paint. As this technology creates new possibilities for creativity and innovation, it also poses ethical questions about originality, authorship, and the place of artificial intelligence in the creative process [9]. In tasks where, for example, students are expected to draw or paint, relying on AI tools is likely to result in reduced effort, missed learning outcomes, and a higher likelihood of cheating.

Art students are well-placed to lead the response to these ethical challenges. With AI becoming an integral part of the creative process, students have to face questions about the originality of AI-assisted work and the possibility of plagiarism [10]. For instance, an art student might use AI to produce visual concepts or mimic a specific style, blurring the line between legitimate help and academic dishonesty. Art students must now balance traditional artistic skills with AI-generated content in a way that maintains academic integrity [11]. For this reason,

these students are often the first to contend with ethical issues pertaining to AI, making them an important group for studying the effects of generative AI on academic behavior [12].

There is currently a lack of research into factors associated with the use of AI technologies, especially in art education. Art students, who are already under pressure to produce unique and creative work, may experience heightened anxiety as they attempt to balance their own skills with AI-generated content [13]. This anxiety could manifest in procrastination, academic dishonesty, and overreliance on AI tools [14]. For example, students might feel inadequate compared to the output produced by AI or be worried that their skills are becoming outdated. The temptation to use AI as a shortcut could then increase, making unethical academic behavior more likely.

In this study, we intend to fill this research gap by focusing on relationships between dark personality traits, academic misconduct, and the use of generative AI use among art students in China. The research investigates how these traits influence academic behaviors in an AI-driven educational environment and sheds light on the ethical and psychological challenges of art students in a broader discussion of the role AI plays in higher education. We propose two research questions:

1. Are the dark personality traits of materialism, narcissism, psychopathy, and Machiavellianism associated with academic dishonesty, anxiety, and procrastination among university art students?
2. Are academic dishonesty, anxiety, and procrastination associated with frustration, negative thinking, and generative AI use among university art students?

This study makes several notable theoretical contributions. It builds upon previous theoretical frameworks [e.g. 15, 16] by establishing connections between negative characteristics and problematic academic behaviors. It also shows how these behaviors are associated with frustration, negative thinking, and reliance on generative AI tools. Moreover, the study situates these associations within the distinct experiences faced by art students in China, addressing knowledge gaps regarding the expression of negative characteristics and their repercussions in this understudied cultural context. The study's findings may inform theoretical frameworks that can guide future research, teaching methods, and interventions designed to enhance academic honesty, well-being, and personal development, particularly among art students.

Literature review and conceptual framework

Social cognitive theory

Social cognitive theory (SCT) developed by Bandura [17] posits that human behavior is a dynamic interaction of personal, environmental, and behavioral factors. A central aspect of SCT is reciprocal determinism, whereby behavior is shaped by internal cognitive factors such as personality traits and external influences such as social norms. It includes key concepts of observational learning and self-regulation. Together, these elements provide a way to better understand academic misconduct.

Academic misconduct is a particularly interesting area to apply SCT because students are influenced by both personal and environmental factors. Students might imitate dishonest behavior if it is rewarded. In stressful environments, weak self-regulation can lead to cheating [8]. Academic misconduct among art students could involve using generative AI tools to create artwork and claiming this is acceptable because everyone else does it or because they are under pressure to succeed. In support of SCT is the idea that behavior is shaped by perceived consequences and peer behavior (Bandura, 1986). According to Rodrigues et al. [18], multiple individual and contextual factors influence the learning process. SCT's idea of a reciprocal nature between behavior and environment makes it a perfect tool for understanding misconduct. For instance, a student's dishonesty can reduce peer group ethical standards, reinforcing future misconduct. This dynamic is especially important in art education where norms surrounding technology use are changing very quickly.

SCT can also be applied to dark personality traits such as materialism, narcissism, Machiavellianism, and psychopathy [19], which are based on self-interest and manipulation. People high in these traits tend to engage in misconduct more if they think that somebody else has benefited from dishonest behavior. For example, a high-Machiavellianism student may rationalize the use of AI to attain higher grades, while a student scoring high in narcissism might cheat out of a sense of entitlement or preserve an image of being a top performer [8]. These examples fit with SCT's notion of self-regulation failure, meaning self-interest will trump moral standards. There may be a neurological basis to such responses, with research indicating distinct neurobiological profiles and mirror neuron functioning among those high in dark triad traits. SCT offers a comprehensive account of how personality traits interact with the external environment to influence behavior. This study attempts to explore the darker side of art students's behavior using SCT, examining how dark personality traits and the academic environment drive dishonest behavior.

Overview of existing knowledge

Although there has been extensive research on dark personality traits and academic misconduct, much of this research has focused on the general student population, not specific groups such as art students. It is well-established that academic dishonesty is related to traits such as narcissism, Machiavellianism, and psychopathy [19]. Many studies, however, neglect context-specific factors such as the pressures of other disciplines. For example, Curtis [20] discovered that people high in Machiavellian tendencies were more likely to commit premeditated cheating, while those high in psychopathy were more likely to be impulsive and dishonest.

Studies into the dark triad and academic dishonesty have largely revolved around text-based assignments, but art students have difficulties with creativity and originality. AI tools might tempt students to improve their work by using the technology, which presents an ethical dilemma in fields that rely on artistic innovation. These students must frequently produce innovative work, elevating the pressure to be creative and original [21]. This unique context implies that dark traits might manifest differently in art students.

Studies of dark personality traits tend to treat these traits as static predictors of behavior without exploring fully the role of environmental factors. Rodrigues et al. [18] recognized academic misconduct as complex, influenced by both personal and environmental factors, but did not investigate whether these dynamics differed across disciplines. For example, art students, who receive highly subjective grades and assessments are more prone to being under external pressures than those in standardized fields. Students can manipulate their environment to succeed and intense competition for recognition may amplify traits such as narcissism or Machiavellianism. Dark personality traits are therefore useful for understanding academic dishonesty, though should be viewed in the context of the pressures and challenges of the academic environment.

Dark personality traits and academic malpractices

Dark Personality Traits are theoretically grounded on the Dark Triad model, which includes traditionally narcissism, Machiavellianism and psychopathy, and is extended to materialism as another trait [19, 22]. Grandiosity, entitlement, and a need for admiration characterize narcissism, while Machiavellianism encompasses manipulative, exploitative behavior that serves the needs of the individual to the detriment of others. Psychopathy is characterized by impulsivity, a lack of empathy, and anti-social tendencies, and materialism refers to an orientation that prioritizes the possession of physical goods and wealth for personal satisfaction. These traits are linked by a core tendency to be self-interested, manipulative, and

disregard social norms. They are associated with many forms of unethical behavior, including academic misconduct [19]. While, materialism can be defined as an individual's overwhelming need for material things, wealth, and external status symbols in order to enhance sense of pleasure and self-value [22]. Students, driven by materialism as a means of personal satisfaction, and therefore might engage in misconduct for extrinsic rewards (such as grades or accolades).

People who exhibit dark traits may demonstrate self-absorption together with manipulation abilities and weak empathy which makes them more likely to pursue activities that benefit their interests regardless of ethical considerations. According to Mace [23], dark psychology can help explain academic dishonesty. Dark personality traits appear to cultivate an attitude that finds dishonest conduct acceptable [24]. Narcissism is associated with a greater likelihood of engaging in negative academic behaviors [19, 20]. There is some evidence of possible mechanisms underlying this association. Grubbs and Exline [24] found that trait entitlement, a characteristic of narcissism, increased vulnerability to psychological distress such as frustration and negative thinking, in turn increasing the likelihood of cheating.

A considerable number of studies have examined the influence of dark traits on academic anxiety. For instance, Sabouri et al. [25] revealed a positive correlation between psychopathy and academic anxiety in university students, while Shengbo et al. [26] found positive correlations between both Machiavellianism and psychopathy and academic anxiety. Psychopathy is characterized by impulsive and reckless behavior, which may generate heightened anxiety when such behavior impacts scholastic responsibilities and leads to missed deadlines. The mechanisms underlying these associations may involve a lack of empathy and a proclivity to disregard societal rules and standards. Individuals high in psychopathy and Machiavellianism may experience challenges in forming positive connections with classmates and teachers, which would likely contribute to elevated academic anxiety.

Moreover, there is compelling evidence supporting the link between dark personality traits and academic dishonesty. Individuals high in dark traits are more inclined to engage in deceptive and manipulative behaviors [27, 28]. Baran and Jonason [29] found psychopathy to be a significant contributor to academic dishonesty, while Curtis [20] found that individuals with darker personality features were more likely to engage in unethical academic conduct.

Researchers have also studied relationships between dark personality traits and academic procrastination. According to Yockey [30], those high in dark traits and a propensity for negative thinking tend to delay tasks due to perfectionist ideals or fear of failing. Individuals

with darker personalities appear more likely to engage in activities that obstruct their ability to succeed in education. Based on the evidence reviewed, we propose the following hypotheses:

H1. Dark personality traits are significantly associated with academic dishonesty.

H2. Dark personality traits are significantly associated with academic anxiety.

H3. Dark personality traits are significantly associated with procrastination.

Academic misconduct and frustration

Academic misconduct includes cheating, plagiarism, fabrication of data, unauthorized collaboration, and misrepresentation of one's work [5]. These behaviors corrupt the reliability of academic achievements and the fair evaluation of students' capacity. According to SCT, personal, environmental, and behavioral factors interact to influence behavior [18]. Students who experience academic anxiety may find themselves more frustrated overall. According to a study by Chi and Qian [31], frustration and anxiety in academic environments are positively associated. Students who generally experience excessive anxiety about their academic performance are more prone to feeling disappointed when faced with difficult assignments or failure. Disappointment can, in turn, produce greater anxiety, starting a vicious cycle that affects well-being and academic performance [32].

Academic dishonesty and misconduct can be viewed as responses to environmental pressures [20, 33]. Individual differences in the level of frustration experienced in reaction to such pressures may help explain why some students cheat and procrastinate and others do not. Nussbaum [34] and Oravec [35] note that academic frustration can be systemic and social. Several authors have acknowledged reciprocal determinism effects, whereby frustration results from personal factors such as perfectionism and negative thinking interacting with behavior (e.g., procrastination) and environmental feedback (e.g., academic demands) [15, 16, 36]. Wiegand [37] found that misalignment of personal expectations and actual behaviors causes negative emotions such as frustration. Procrastination is an example of misaligned expectation (achieving good results) and behavior (not completing tasks) [38]. According to the SCT's triadic reciprocity concept, students may cheat to cope with systemic pressures, which frustrates them when their actions violate internal standards or have negative consequences. Based on this evidence, we predict that:

H4. Academic dishonesty, anxiety, and procrastination are significantly associated with frustration.

Academic misconduct and negative thinking

SCT suggests that cognitive factors shape behavior and psychological states. As such, academic misconduct may lead to negative thinking. Ehring et al. [15] reported that repetitive negative thinking is content-independent, suggesting that academic dishonesty and anxiety may cause students to think negatively. Academically anxious students frequently engage in catastrophizing and negative self-talk, and tend to have illogical ideas about their performance and talent in the classroom [39]. These maladaptive mental patterns can exacerbate anxiety and erode self-assurance, further affecting performance. In addition, Macedo et al. [16] found that repetitive negative thinking mediated the relationship between perfectionism and psychological disorders, suggesting that harmful patterns of thought such as perfectionism can have serious health consequences.

Dark triad traits predispose students to negative thinking patterns as they are linked to a tendency to rationalize dishonest behavior, according to Jones and Paulhus [19] and Curtis et al. [20]. Academic misconduct can have immediate behavioral consequences and long-term negative effects, perpetuating a cycle of negative thinking. Therefore, we predict that:

H5. Academic dishonesty, anxiety, and procrastination are significantly associated with negative thinking.

Academic misconduct and generative AI use habits

In SCT, observational learning and behavioral reinforcement help shape behavior [18, 40]. Students who commit academic dishonesty and experience academic anxiety may use generative AI tools and related tools more frequently to complete their assignments, thus becoming more dependent on these technologies. Such behavior

could be reinforced if the outcomes are favorable, such as receiving good grades with less effort [17, 24]. According to Toribio [41], university students who experience higher levels of academic anxiety tend to use online learning resources more frequently. More anxious students may be more likely to seek extra help and resources and turn to AI assistants to reduce their concerns.

Personality traits might impact this relationship [14]. Research into dark triad traits by Jones and Paulhus [19] indicates that individuals with more pronounced dark traits are more likely to take advantage of resources, while Baran and Jonason [29] found that students high in psychopathy were more likely to engage in misconduct to further their objectives. Moreover, research on social media use by Dhir and Tsai [14] shows that technology is frequently used as a coping strategy to procrastinate. In academic settings, this could manifest as a propensity to use generative AI, as procrastinators look for quick fixes to meet deadlines. We therefore speculate a link between academic misconduct and increased generative AI use habits:

H6. Academic dishonesty, anxiety, and procrastination are significantly associated with generative AI use habits.

To summarize, we predict that the dark personality traits of materialism, narcissism, psychopathy, and Machiavellianism are linked to variables that influence academic misconduct, namely dishonesty, anxiety, and procrastination. In turn, these variables related to academic misconduct are predicted to be positively associated with higher frustration, negative thinking, and generative AI use among university art students. Figure 1 presents our conceptual framework and predicted associations. According to the framework, darker traits influence problematic academic behavior and anxiety, which

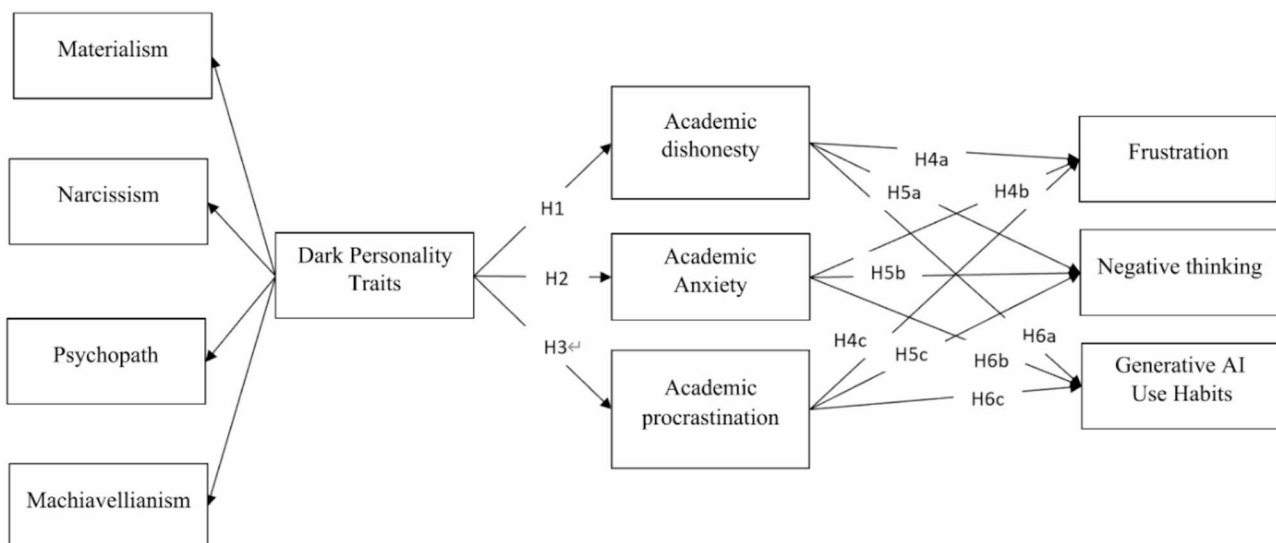


Fig. 1 Conceptual Framework

in turn affect emotional and cognitive states and reliance on generative AI tools.

Methods

Participants and procedure

A stratified random sampling method was used to recruit university art students from different prominent institutions in Sichuan province, China, to ensure diversity of disciplines. The universities chosen are *Sichuan University of Media and Communications*, *Sichuan Conservatory of Music*, *Sichuan University of Culture and Arts*, *Sichuan Vocational College of Art*, *Sichuan Film and Television University* and *Sichuan Fine Arts Institute* because of their dance, drama, music and visual arts art programs. These are institutions that are well known for their high artistic and creative competencies of their graduates, as well as their diverse student population and curricula that integrate traditional techniques with modern technologies like AI tools.

From the enrolment data of the six participating universities ($\approx 22,500$ total art students), we calculated the proportional representation targets for our sample of 750 students (500 in person, 250 online). To accommodate the variations at both the university level and discipline level in Sichuan’s art education landscape, the sampling process was structured in multiple stages.

The target population was first divided into mutually exclusive strata according to two key characteristics: (1) institutional affiliation across the six universities representing approximately 18–22% of Sichuan’s total art student population and (2) the four major art disciplines (dance = 15%, drama/theater = 25%, music = 35%, and visual arts = 25% of the art student population). The dual stratification guaranteed a balanced representation in both institutional and artistic specializations.

We did a pilot test with 30 participants before giving the main survey to ensure the clarity and validity of the questionnaire. The feedback from this pilot phase aided in identifying ambiguities in wording and structure that

required adjustment in the final survey instrument. This step guaranteed that the questions were well understood and effectively captured the responses that were intended.

We received 555 completed responses (350 in person and 205 through the web) from 750 distributed surveys, which gave us a 74% response rate. We excluded 51 responses due to incomplete or inconsistent data and the final sample was 504 valid responses. The dataset was thus cleaned rigorously to ensure its reliability for further analysis.

As shown in Tables 1 and 41% of participants were female and 59% were male. Students from different art disciplines were well-represented, with 25% studying visual arts, 36% drama, 26% music, and 13% dance. The distribution of participants across disciplines was largely in line with the composition of students on the universities’ art programs, suggesting that the sample was representative of the broader student population enrolled in these fields.

Measures

We used established and validated scales to measure the variables of interest. These scales had been thoroughly tested in previous studies, thus increasing the robustness of our findings and the ability to compare findings with those of similar studies. Where needed, the scales were slightly adapted to fit the context of the research while preserving the meaning of the scales.

We adapted and translated internationally validated scales from previous research into Chinese for our questionnaire development. A rigorous translation followed by back-translation process ensured both linguistic precision and cultural relevancy of the scales. The original English language scales underwent translation into Chinese by an expert bilingual individual. An independent bilingual translator conducted the English version of the Chinese text through back-translation. A group of academic experts including professors and PhD students evaluated the final translated scales for content validity together with their clarity and contextual appropriateness before suggesting any needed improvements. Consequently, we ensured the face validity of all constructs. In order to make it more rigorous, a pilot study was done with 30 students. The reliability values for all scales adapted were well above the threshold of 0.7. Which allowed us to use scales in main survey. The detail of original scales is given below.

Dark personality traits: materialism, machiavellianism, narcissism, and psychopathy

We used the Materialistic Possession subscale from the Chinese Adolescent Materialism Scale [CAMS; 22] to measure materialism. Although this scale was tested on

Table 1 Demographic characteristics

Variable	Response	Frequency (N = 504)	%
Gender	Male	296	58.7%
	Female	208	41.3%
Subject	Dance	65	12.9%
	Drama	181	35.9%
	Music	131	26.0%
	Visual Art	127	25.2%
Location	Rural	260	51.6%
	Urban	244	48.4%
Semester	1st	35	6.9%
	2nd	104	20.6%
	3rd	156	31.0%
	4th	209	41.5%

adolescent, Shek et al. [22] determined that The CAMS is a reliable and valid instrument to measure adolescent materialism in Hong Kong. This research also provides initial youth profile and psychological-social factors of materialism in Hong Kong. This is relevant to the public concern of materialism in young people, and provides a useful measure for future scientific studies in Chinese populations. Similarly, the majority of students targeted in this study belong to youth category which justified the use of CAMS. Secondly, the materialistic possession dimension of CAMS included very generic questions which can be generalized to adult groups. Finally, we performed the initial pilot testing which helped us to assess the reliability in university context. In the present study, the scale showed good internal consistency with a Cronbach's α value of 0.87.

We employed the Short Dark Triad scale [SD3; 19] to measure the three traditional dark triad traits. The scale consists of 27 items, with three subscales of nine items each to assess the three traits. Jones and Paulhus [19] identified a three-factor structure, with each subscale demonstrating acceptable internal consistency ($\alpha = 0.68$ – 0.78), and observed strong correlations with longer dark triad measures in expected directions (Mach-IV, NPI, and SRP-III), indicating the SD3's effectiveness as a brief instrument. In the present study, internal consistency was excellent (Cronbach's $\alpha = 0.95$).

A Likert scale was used to record responses for the dark personality traits.

Academic dishonesty

Participants completed the Academic Dishonesty Scale [42, 43]. The scale consists of 10 items that ask participants to report their frequency of engaging in academically dishonest behaviors since beginning their college careers on a Likert scale. Example item is "Copied material and turned it in as your own work". In the present study, the scale's internal consistency was excellent (Cronbach's $\alpha = 0.95$).

Academic anxiety

The Academic Anxiety Scale, adapted by following the studies of Wang et al. [44] and Cassady et al. [45], consists of eight items. The responses were gathered on a Likert scale. Internal consistency was excellent (Cronbach's $\alpha = 0.95$).

Academic procrastination

The short form of the Academic Procrastination Scale [APS-S; 30, 46] is a 5-item measure. Responses were gathered on a Likert scale. Internal consistency was excellent (Cronbach's $\alpha = 0.92$).

Frustration

We used the Frustration scale adapted from [47] to assess school-based frustrations. Responses were recorded on a 5-point Likert scale. Internal consistency was good (Cronbach's $\alpha = 0.87$).

Negative thinking

We used the 5-item Negative Thinking subscale from the Persistent and Intrusive Negative Thoughts Scale [PINTS; 48]. Respondents are asked to rate the frequency of certain aspects of negative thinking on a likert scale. Internal consistency was good (Cronbach's $\alpha = 0.86$).

Generative AI use habit

We used a modified version of the Habit subscale, previously used by Strzelecki [49] and adapted from Venkatesh and Xu [50]. The measure has four items. The responses were gathered on a Likert scale. In Strzelecki's [49] study, the subscale demonstrated good reliability and convergent validity, with a composite reliability of 0.92 and an AVE value of 0.73. In the present study, internal consistency was good (Cronbach's $\alpha = 0.88$).

Common method bias

Several statistical approaches were used to account for issues with common method bias, including the partial correlation method and Harman's single-factor test [51]. Using Harman's single-factor test, all survey item responses were loaded onto a singular factor, which accounted for 19% of the overall variance. This value was significantly below 50%, indicating common method bias was not a cause for concern. Partial correlation analysis revealed that the inclusion of the marker variable in the model did not have any impact on projected routes or their significance. Moreover, as correlation coefficients were all below $r = .7$, there were no apparent multicollinearity issues (Table 3).

Results

Confirmatory factor analysis

We employed structural equation modeling (SEM) using AMOS software to assess validity and test hypothesized paths. Given the reflective nature of the model, covariance-based SEM (CB-SEM) was judged appropriate [52]. AMOS is commonly used to conduct CB-SEM given its capability to handle complex models. The first step in SEM is confirmatory factor analysis (CFA), which is conducted by developing a measurement model to examine the convergent and discriminant validity of constructs. During model development, all constructs were considered first-order variables except dark personality, which was treated as a second-order construct with four dimensions at the first order. Both theoretical and empirical evidence supported treating dark personality

traits—narcissism, Machiavellianism, and psychopathy—as a second-order construct. These traits are separable, yet possess common features such as manipulation, self-interest, and lack of empathy (Paulhus & Williams, 2002) which allows them to be conceptualized as facets of a general tendency to engage in socially aversive behavior. This approach was confirmed by confirmatory factor analysis, with each trait loading on a higher-order dark personality factor, consistent with research showing high correlations among the traits (Jones & Paulhus, 2014). Model fit was examined using $\chi^2/df < 3$, CFI > 0.90, TLI > 0.90, and RMSEA < 0.08 [52]. Findings showed no concerns related to model fit ($\chi^2/df = 1.97$, CFI = 0.96, TLI = 0.92, and RMSEA = 0.04).

We then assessed convergent validity. Table 2 presents our assessment of convergent validity based on the thresholds of Hair Jr. et al. [53] and Ringle and Sarstedt [54]. The key indicators are factor loadings, composite reliability, and average variance extracted (AVE). Convergent validity is satisfied if factor loadings for each item are greater than 0.5, AVE is greater than 0.5, and composite reliability is greater than 0.8. Table 2 shows that all constructs met these convergent validity thresholds.

The model's discriminant validity was tested using Fornell and Larcker's framework [55] which requires the square root of AVE values from each construct to exceed correlations between constructs. Table 3 demonstrates that discriminant validity exists because square roots of the AVEs exceed their respective correlations.

Assessment of path model

We next developed a structural model, applying CB-SEM to test the research hypotheses (Table 4; Fig. 2). Model fit was ensured as all indices were in line with recommended cutoffs ($\chi^2/df = 2.11$, CFI = 0.91, TLI = 0.91, and RMSEA = 0.047). The path analysis, presented in Table 4, indicated significant associations between dark personality traits and academic dishonesty ($\beta = 3.985$), academic anxiety ($\beta = 4.769$), and academic procrastination ($\beta = 2.893$), with p -values all < 0.05. Hypotheses H1, H2, and H3 were thus supported. Students with high psychopathy, Machiavellianism, and narcissism were more likely to report engaging in unethical academic practices, experiencing elevated anxiety in academic settings, and procrastinating on assignments.

Frustration was significantly positively associated with academic dishonesty ($\beta = 0.158$), academic anxiety ($\beta = 0.165$), and academic procrastination ($\beta = 0.102$), supporting H4a–c. Negative thinking was also positively linked with academic dishonesty ($\beta = 0.127$), academic anxiety ($\beta = 0.092$), and academic procrastination ($\beta = 0.112$), supporting H5a–c. Lastly, generative AI use was significantly associated with academic dishonesty ($\beta = 0.195$), academic anxiety ($\beta = 0.094$), and academic

procrastination ($\beta = 0.192$), supporting H6a–c. Overall, the analysis indicated that those who engaged in unethical academic practices experienced heightened academic anxiety, procrastinated more, and were more likely to rely on generative AI tools.

Besides, we also tested the effect size of the coefficients by calculating the Cohen's effect size in AMOS. In this regard, we calculate the squared multiple correlations (R^2) of each relationship in the model. Afterwards, we compared combined R^2 with individual R^2 . The results are given in Table 5. It shows that dark personality traits have a greater effect on academic anxiety as compared to academic dishonesty and academic procrastination. While, in relation to frustration, academic anxiety has slight edge over academic dishonesty and academic procrastination. However, academic dishonesty has great effect on negative thinking as compared to academic anxiety and academic procrastination. Finally, academic dishonesty has great effect on generative AI use as compared to academic anxiety and academic procrastination.

Overall, Table 5 shows that dark personality traits have moderate effect on academic anxiety and academic dishonestly. Rest, all effects in the model are weak.

Discussion

The research investigates how dark personality traits affect academic misconduct and AI tool utilization among Chinese art students. The findings show that dark personality traits including narcissism, machiavellianism, psychopathy and materialism are directly associated with academic dishonesty and anxiety and procrastination behaviors. The behaviors led to higher levels of frustration, negative thinking and dependency on generative AI tools. We discuss these findings in depth using relevant research to explain how these relationships work as well as their impact on art educational practices.

Firstly, the study results are consistent with SCT in which personal dispositions, behavior, and environmental dimensions are viewed as mutually influential [17]. For instance, the high correlations between dark personality traits and academic misconduct demonstrate SCT's basic point that personal dispositions (narcissism or Machiavellianism, for example) along with environmental conditions (academic competitiveness or AI availability, for example) influence behavior. The findings also show how anxiety and procrastination serve as mediators that help to support SCT's assertion that self-regulation has a significant effect on ethical decision-making. The increasing use of generative AI tools by students with more negative thinking and frustration corroborates the manner in which environmental resources (like AI) and personal attitudes interactively affect behavior in academic settings. Such results emphasize how SCT is still relevant to

Table 2 Convergent validity and reliability tests

Scale	Items	Factor Loading	Composite Reliability	AVE
Academic Dishonesty (AD)	AD1	0.884	0.951	0.662
	AD2	0.932		
	AD3	0.921		
	AD4	0.859		
	AD5	0.871		
	AD6	0.800		
	AD7	0.688		
	AD8	0.692		
	AD9	0.713		
	AD10	0.723		
Academic Anxiety (AAN)	AAN1	0.857	0.959	0.744
	AAN2	0.863		
	AAN3	0.810		
	AAN4	0.870		
	AAN5	0.866		
	AAN6	0.861		
	AAN7	0.889		
	AAN8	0.883		
Academic Procrastination (AP)	AP1	0.877	0.919	0.696
	AP2	0.855		
	AP3	0.861		
	AP4	0.815		
	AP5	0.758		
Generative AI Use Habit (GAUH)	CAUH1	0.802	0.885	0.659
	CAUH2	0.866		
	CAUH3	0.789		
	CAUH4	0.787		
Frustration (FRUS)	FRUS1	0.773	0.874	0.634
	FRUS2	0.777		
	FRUS3	0.856		
	FRUS4	0.776		
Negative Thinking (NT)	NT1	0.703	0.862	0.555
	NT2	0.738		
	NT3	0.745		
	NT4	0.764		
	NT5	0.772		
Machiavellianism (MACH)	MACH1	0.838	0.954	0.697
	MACH2	0.855		
	MACH3	0.840		
	MACH4	0.822		
	MACH5	0.840		
	MACH6	0.865		
	MACH7	0.839		
	MACH8	0.815		
	MACH9	0.800		
Materialism (MAT)	MAT1	0.856	0.931	0.731
	MAT2	0.866		
	MAT3	0.844		
	MAT4	0.857		
	MAT5	0.852		

Table 2 (continued)

Scale	Items	Factor Loading	Composite Reliability	AVE
Narcissism (NAR)	NAR1	0.826	0.940	0.634
	NAR2	0.806		
	NAR3	0.849		
	NAR4	0.700		
	NAR5	0.769		
	NAR6	0.836		
	NAR7	0.818		
	NAR8	0.786		
	NAR9	0.765		
Psychopathy (PSY)	PSY1	0.785	0.948	0.671
	PSY2	0.835		
	PSY3	0.806		
	PSY4	0.808		
	PSY5	0.807		
	PSY6	0.802		
	PSY7	0.820		
	PSY8	0.854		
	PSY9	0.854		

Table 3 Discriminant validity tests

	PSY	MAT	MACH	AAN	NAR	AP	CAUH	NT	FRUS	AD
PSY	0.819									
MAT	0.128	0.855								
MACH	0.382	0.048	0.835							
AAN	0.396	0.130	0.252	0.863						
NAR	0.490	0.035	0.493	0.400	0.796					
AP	0.249	0.052	0.205	0.136	0.271	0.834				
GAUH	0.196	0.029	0.139	0.199	0.230	0.248	0.812			
NT	0.177	0.066	0.133	0.166	0.209	0.156	0.241	0.745		
FRUS	0.355	0.066	0.326	0.263	0.687	0.150	0.207	0.310	0.796	
AD	0.299	0.053	0.250	0.236	0.320	0.041	0.284	0.202	0.259	0.814

Note: Square Root of AVEs are given in diagonal with bold highlighted numbers

Table 4 Path analysis

Path	B	t	p
H1. Dark Personality Traits $\diamond$ Academic dishonesty	0.413	2.047	0.041
H2. Dark Personality Traits $\diamond$ Academic anxiety	0.530	2.073	0.038
H3. Dark Personality Traits $\diamond$ Academic procrastination	0.335	2.009	0.045
H4a. Academic dishonesty $\diamond$ Frustration	0.210	4.39	< 0.001
H4b. Academic anxiety $\diamond$ Frustration	0.165	4.277	< 0.001
H4c. Academic procrastination $\diamond$ Frustration	0.208	2.573	0.01
H5a. Academic dishonesty $\diamond$ Negative thinking	0.171	3.445	< 0.001
H5b. Academic anxiety $\diamond$ Negative thinking	0.117	2.343	0.019
H5c. Academic procrastination $\diamond$ Negative thinking	0.138	2.76	0.006
H6a. Academic dishonesty $\diamond$ Generative AI use	0.247	5.231	< 0.001
H6b. Academic anxiety $\diamond$ Generative AI use	0.113	2.402	0.016
H6c. Academic procrastination $\diamond$ Generative AI use	0.208	4.677	< 0.001

understanding unethical academic behavior, particularly in dynamic learning environments.

The study shows that students who display more dark personality characteristics tend to perform unethical academic work while also experiencing anxiety about their studies and delaying their tasks. Narcissistic people engage in dishonest conduct to sustain their self-image and get recognition because they seek constant admiration and validation. Students who are Machiavellian demonstrate manipulative traits and sceptical attitudes toward others and view academic misconduct as an instrumental method to reach their academic goals. Psychopathic individuals demonstrate impulsive behavior coupled with their inability to empathize so they disregard ethical standards to cheat without showing remorse. The current research supports earlier Dark Triad research which established a direct relationship between these behavioral characteristics and unethical conduct and self-control breakdowns [19, 29].

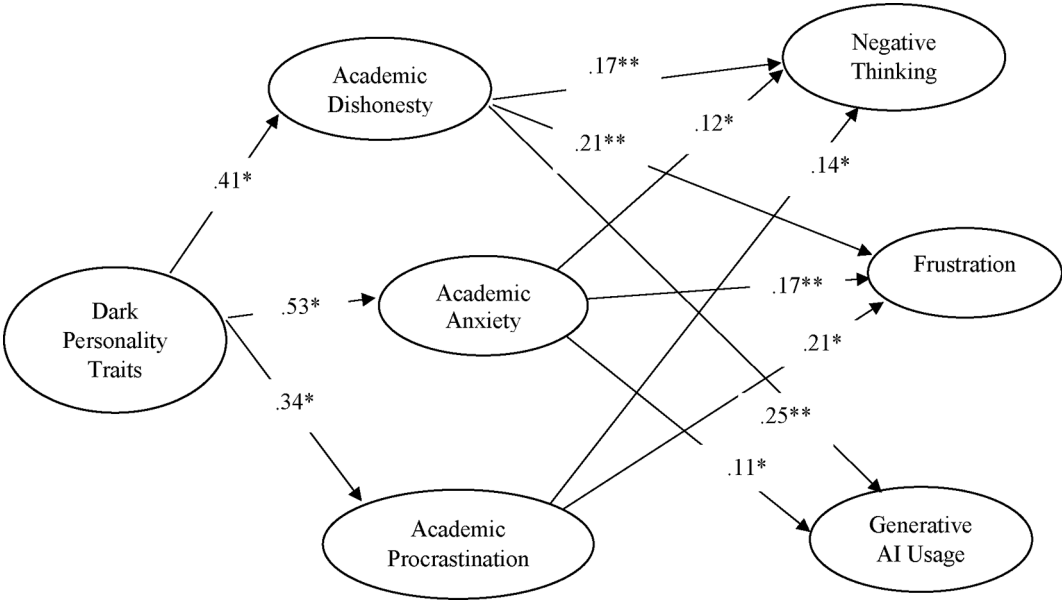


Fig. 2 Structural Model with Estimates. Note: *p-value < 0.05, **p-value < 0.00

Table 5 Effect size (Cohen's f^2)

Independent Variable	Dependent Variable	R_E^2	R_I^2	$(R_I^2 - R_E^2)$	$(1 - R_I^2)$	$F^2 = (R_I^2 - R_E^2) / (1 - R_E^2)$
Dark Personality Traits	Academic dishonesty	N/A	0.175	N/A	N/A	0.175
Dark Personality Traits	Academic anxiety	N/A	0.277	N/A	N/A	0.277
Dark Personality Traits	Academic procrastination	N/A	0.105	N/A	N/A	0.105
Academic dishonesty	Frustration	0.114	0.150	0.036	0.850	0.042
Academic anxiety	Frustration	0.103	0.150	0.047	0.850	0.055
Academic procrastination	Frustration	0.125	0.150	0.025	0.850	0.029
Academic dishonesty	Negative thinking	0.058	0.093	0.035	0.907	0.037
Academic anxiety	Negative thinking	0.080	0.093	0.013	0.907	0.014
Academic procrastination	Negative thinking	0.068	0.093	0.025	0.907	0.027
Academic dishonesty	Generative AI use	0.095	0.162	0.067	0.838	0.074
Academic anxiety	Generative AI use	0.151	0.162	0.011	0.838	0.013
Academic procrastination	Generative AI use	0.110	0.162	0.052	0.838	0.062

Research results demonstrate that academic misconduct and both anxiety and procrastination tendencies lead to higher levels of frustration and negative thought patterns. Students who perform dishonest activities face frustration because their external actions oppose their personal moral standards and because they encounter negative repercussions. Students who experience academic anxiety because of perfectionism or fear of failure tend to develop recurring negative thoughts that include catastrophic thinking and self-critical thinking. Student procrastination as an expectation-behavior mismatch leads to the intensification of negative emotions which forms a continuous destructive pattern that harms academic success and quality of life. The findings line up with research about academic dishonesty which demonstrates that unethical actions produce frustration and negative thoughts as well as unhelpful coping methods [15, 16].

Importantly, the current study determines that academic misconduct in terms of academic misconduct, academic procrastination, and academic anxiety shows a direct correlation with the growing dependence on generative AI tools. Academic students who practice dishonest conduct seek AI tools because they want to circumvent difficult work or get quick solutions particularly under high academic pressure or subjective assessment conditions. AI tools serve as a stress-reducing mechanism for students who feel academic anxiety and procrastinators use them to perform their assignments just before deadlines. Student perception of AI tools as success shortcuts leads them to continue unethical practices while ignoring the ethical consequences. Research on educational technology demonstrates how AI functions as both a resource and an ethical enabler of wrong behaviors [14, 41].

The study has investigated dark personality traits in academic misconduct but we should note that positive traits including creative thinking function as protective measures against academic dishonesty. Creative thinking enables students to handle academic tasks through innovative methods which reduces their need to use dishonest practices like plagiarism along with improper usage of generative AI tools [56]. The problem-solving abilities and inner drive that emerge from creative thinking decrease students' behavior of academic dishonesty. Students who receive creative thinking encouragement show enhanced work engagement through original thinking which leads them to avoid unethical practices like cheating and plagiarism [57].

Managerial and university implications

The results of this study have implications for administrators at universities and legislators interested in academic integrity and student welfare. First, the strong relationship between dark personality traits and academic anxiety implies that universities should take proactive steps to recognize and assist students who may be at risk. This could be accomplished through initiatives to strengthen moral judgment and emphasize the value of academic integrity, perhaps through the implementation of honor codes with repercussions for misbehavior. School-based psychological services could help to address underlying problems causing student misconduct, for instance by teaching coping mechanisms to manage frustration and negative thoughts.

Second, the evidence might inspire inventive teaching methods. Redesigning assessment procedures and curricula may be necessary to enhance personalization, application-oriented approaches, and cultivation of higher-order abilities that are less susceptible to plagiarism and which artificial intelligence cannot accurately mimic. Integrating lessons regarding the appropriate use of AI into the academic curriculum could enhance student understanding of its associated hazards and ethical quandaries. Such forward-thinking learning designs have the potential to reduce academic frustration and thus decrease the inclination to depend on AI.

As generative AI use is linked to dishonesty and anxiety, educational institutions could monitor student usage patterns on digital platforms to identify possible psychological and academic problems and provide targeted support when needed. Awareness of interactions among personality traits, psychological distress, and academic behavior could help universities promote a more comprehensive approach to student development and integrity.

Limitations and future directions

This study's primary limitation was its sole reliance on self-reported data, which can be influenced by social

desirability bias resulting in the underreporting of undesirable characteristics and behaviors. Such effects could be minimized in future studies by employing objective metrics for academic dishonesty and a multi-method approach that includes using observational and qualitative data to evaluate personality traits. Further, the study's cross-sectional design restricts the capacity to assume causal relationships. Longitudinal studies would be helpful to learn how relationships between dark traits and academic behaviors change over time.

The sample comprised only art students, which impedes generalizability of the findings. Art students may have unique experiences, creative practices, and characteristics that influence links between dark personality traits, academic behaviors, and psychological impacts. As such, caution is recommended when extrapolating the results to students in other fields. Relatedly, the results were derived from a specific academic setting and may not apply to other educational environments or cultural contexts. It would be beneficial to repeat the study in other settings to strengthen the validity of the results.

The study only explored AI generative tools in a general sense rather than considering specific tools such as Midjourney or ChatGPT. Future study is recommended to explore these tools in relation to students' dark personalities. Finally, extending the scope of the study to include positive personality traits and their influence on academic integrity could inform a more balanced perspective of student behavior and provide insights into protective factors that might reduce the likelihood of academic misconduct.

Conclusion

The results of this study of Chinese university art students reveal that elevated expression of dark personality traits is associated with academic dishonesty, anxiety, and procrastination, which are, in turn, linked to increased frustration, negative thinking, and reliance on generative AI tools. Based on these findings, it is suggested that educational institutions implement early identification systems for students who display dark personality and undertake specialized interventions, such as cognitive-behavioral counseling and ethics-based interventions. Academic integrity courses should be part of the curriculum, paying attention both to the consequences of dishonest behavior and the ways of emotional management and self-control. Also, educators should also be trained with the skills to construct assessment methods that minimize the danger of misconduct, for example, by creating process-oriented tasks that will limit the use of AI just before deadlines.

Finally, educational bodies should provide detailed rules and monitoring procedures for AI integration in coursework, so students know what is acceptable versus dishonest AI use. Because creative thinking appears to

reduce risks, art courses should emphasize originality, intrinsic motivation, and studio-based instruction as well as frequent reflective critiques. The implementation of these strategies can help to reduce dishonest shortcuts, and engender a student body that is honest and creatively resilient.

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Author contributions

J.S. is the first author of this work and contributed to the planning of the research model, the literature review, the methodology, and the writing of the remaining sections of this manuscript. S.L. is the second author and the corresponding author of this work, responsible for proposing and developing the research hypotheses and reviewing the conclusions.

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Data availability

The dataset used and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Human ethics and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and the Personal Information Protection Law of the People's Republic of China. The study design was reviewed by the Academic Affairs Office of Sichuan University of Media and Communications, which determined that ethical approval was not required due to the cross-sectional design and the absence of experimental procedures. Prior to data collection, all participants were fully informed about the study's objectives, procedures, and potential risks and benefits. They were assured that their participation was voluntary, and that they could withdraw from the study at any time without consequence. Participants were also informed that their data would remain private and confidential, and would only be used for research purposes. Written informed consent was obtained from all participants before they took part in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Sundar SS, Liao M. Calling BS on chatgpt: reflections on AI as a communication source. *J Commun Monogr*. 2023;25:165–80.
- Johnson N, Sarkar S. Integrating AI. Education in small Liberal arts institutions: curriculum and pedagogical considerations. *Int J Cybern Inf*. 2023;12:1.
- Lee VR, Pope D, Miles S, Zárate RC. Cheating in the age of generative AI: A high school survey study of cheating behaviors before and after the release of ChatGPT. *Comput Educ Artif Intell*. 2024;7:100253.
- Almasri F. Exploring the impact of artificial intelligence in teaching and learning of science: A systematic review of empirical research. *Res Sci Educ*. 2024;54:977–97.
- Christensen Hughes J, Eaton SE. In: Eaton SE, Christensen Hughes J, editors. Academic misconduct in higher education: beyond student cheating BT - Academic integrity in Canada: an enduring and essential challenge. Cham: Springer International Publishing; 2022. pp. 81–102.
- Huang R, Liu S, Rutten K. Why does it matter? A literature review on the functions of school arts education in China 2001–2022. *Asia Pac Educ Rev*. 2024;1–14.
- Jinglian W, Guochuan M, Hua X, Hearst N. 120The Role of the Dual-Track System and Its Consequences. Whither China? Restarting the Reform Agenda. 2016;0.
- Cheung YK, Egan V. The HEXACO-60, the dark triad and scholastic cheating. *Psychol Rep*. 2021;124:2774–94.
- Elgammal A, Kang Y, Den Leeuw M. Automated analysis of drawings at the stroke level for attribution and authentication using artificial intelligence. *Analytical chemistry for the study of paintings and the detection of forgeries*. Springer; 2022. pp. 133–54.
- GAFFAR H. ALBARASHDI S. Copyright protection for AI-Generated works: exploring originality and ownership in a digital landscape. *Asian J Int Law*. 2024;1–24.
- Gulumbé BH, Audu SM, Hashim AM. Balancing AI and academic integrity: what are the positions of academic publishers and universities? *AI Soc*. 2024. <https://doi.org/10.1007/s00146-024-01946-8>
- Nguyen A, Ngo HN, Hong Y, Dang B, Nguyen B-PT. Ethical principles for artificial intelligence in education. *Educ Inf Technol*. 2023;28:4221–41.
- Lin H, Chen Q. Artificial intelligence (AI) -integrated educational applications and college students' creativity and academic emotions: students and teachers' perceptions and attitudes. *BMC Psychol*. 2024;12:487.
- Dhir A, Tsai C-C. Understanding the relationship between intensity and gratifications of Facebook use among adolescents and young adults. *Telemat Inf*. 2017;34:350–64.
- Ehring T, Zetsche U, Weidacker K, Wahl K, Schönfeld S, Ehlers A. The perseverative thinking questionnaire (PTQ): validation of a content-independent measure of repetitive negative thinking. *J Behav Ther Exp Psychiatry*. 2011;42:225–32.
- Macedo A, Soares MJ, Amaral AP, Nogueira V, Madeira N, Roque C, et al. Repetitive negative thinking mediates the association between perfectionism and psychological distress. *Pers Individ Dif*. 2015;72:220–4.
- Bandura A. Social foundations of thought and action. Englewood Cliffs, NJ. 1986;1986:2.
- Rodrigues F, Figueiredo N, Jacinto M, Monteiro D, Morouço P. Social-cognitive theories to explain physical activity. *Educ Sci*. 2023;13:122.
- Jones DN, Paulhus DL. Introducing the short dark triad (SD3) a brief measure of dark personality traits. *Assessment*. 2014;21:28–41.
- Curtis GJ. It Kant be all bad: contributions of light and dark triad traits to academic misconduct. *Pers Individ Dif*. 2023;212:112262.
- Watikinnakorn C, Seesai J, Kerdvibulvech C. Blurring the lines: how AI is redefining artistic ownership and copyright. *Discov Artif Intell*. 2023;3:37.
- Shek DTL, Ma CMS, Lin L. The Chinese adolescent materialism scale: psychometric properties and normative profiles. *Int J Disabil Hum Dev*. 2014;13:285–95.
- Mace R. Dark psychology and gaslighting manipulation: influence human behavior with Mind control techniques: how to camouflage, attack and defend yourself. Ryan Mace; 2022.
- Grubbs JB, Exline JJ. Trait entitlement: A cognitive-personality source of vulnerability to psychological distress. *Psychol Bull*. 2016;142:1204.
- Sabouri S, Gerber M, Lemola S, Becker SP, Shamsi M, Shakouri Z, et al. Examining dark triad traits in relation to sleep disturbances, anxiety sensitivity and intolerance of uncertainty in young adults. *Compr Psychiatry*. 2016;68:103–10.
- Shengbo L, Fiaz M, Mughal YH, Wisetsri W, Ullah I, Ren D, et al. Impact of dark triad on anxiety disorder: parallel mediation analysis during pandemic. *Front Psychol*. 2022;13:914328.
- Kaufman SB, Yaden DB, Hyde E, Tsukayama E. The light vs. dark triad of personality: contrasting two very different profiles of human nature. *Front Psychol*. 2019;10:467.
- Sevi B, Urganci B, Sakman E. Who cheats? An examination of light and dark personality traits as predictors of infidelity. *Pers Individ Dif*. 2020;164:110126.
- Baran L, Jonason PK. Academic dishonesty among university students: the roles of the psychopathy, motivation, and self-efficacy. *PLoS ONE*. 2020;15:e0238141.
- Yockey RD. Validation of the short form of the academic procrastination scale. *Psychol Rep*. 2016;118:171–9.
- Chi Z, Qian L. The impact of Chinese college students' perceived stress on anxiety during the COVID-19 epidemic: the mediating role of irrational beliefs. *Front Psychiatry*. 2021;12:731874.

32. Wagner M, Van Phu N, Azomahou T, Wehrmeyer W. The relationship between the environmental and economic performance of firms: an empirical analysis of the European paper industry. *Corp Soc Responsib Environ Manag*. 2002;9:133–46.
33. Esteves GGL, Oliveira LS, de Andrade JM, Menezes MP. Dark triad predicts academic cheating. *Pers Individ Dif*. 2021;171:110513.
34. Nussbaum MC. Anger and forgiveness: resentment, generosity, justice. Oxford University Press; 2016.
35. Oravec JA. The dark side of academics? Emerging issues in the gaming and manipulation of metrics in higher education. *Rev High Educ*. 2019;42:859–77.
36. Cunha MS, da C. The role of perfectionism and repetitive negative thinking in medicine and dentistry students' burnout. 2022.
37. Wiegand JP. When overqualification turns dark: A moderated-mediation model of perceived overqualification, narcissism, frustration, and counterproductive work behavior. *Pers Individ Dif*. 2023;214:112351.
38. Frayon S, Swami V, Wattlez G, Nedjar-Guerre A, Galy O. An examination of procrastination in a multi-ethnic population of adolescents from new Caledonia. *BMC Psychol*. 2023;11:1.
39. Li D, Li W, Lin X, Zhu X. Associations between rejective parenting style and academic anxiety among Chinese high school students: the chain mediation effect of Self-Concept and positive coping style. *Int J Ment Health Promot*. 2025;27.
40. Jusyte A, Schöenberg M. Impaired social cognition in violent offenders: perceptual deficit or cognitive bias? *Eur Arch Psychiatry Clin Neurosci*. 2017;267:257–66.
41. Toribio NF. Analysis of Chatgpt and other ai's ability to reduce anxiety of Science-Oriented learners in academic engagements. *J Namibian Stud Hist Polit Cult*. 2023;33:5320–37.
42. Bolin AU. Self-control, perceived opportunity, and attitudes as predictors of academic dishonesty. *J Psychol*. 2004;138:101–14.
43. McCabe DL, Trevino LK. Individual and contextual influences on academic dishonesty: A multicampus investigation. *Res High Educ*. 1997;38:379–96.
44. Wang F, Yang Y, Cui T. Development and validation of an academic Involvement scale for college students in China. *Psychol Sch*. 2024;61:847–60.
45. Cassidy JC, Pierson EE, Starling JM. Predicting student depression with measures of general and academic anxieties. *Frontiers in education*. Frontiers Media SA; 2019. p. 11.
46. McCloskey J. Finally, my thesis on academic procrastination. 2012.
47. Buzzai C, Sorrenti L, Costa S, Toffle ME, Filippello P. The relationship between school-basic psychological need satisfaction and frustration, academic engagement and academic achievement. *Sch Psychol Int*. 2021;42:497–519.
48. Magson NR, Rapee RM, Fardouly J, Forbes MK, Richardson CE, Johnco CJ, et al. Measuring repetitive negative thinking: development and validation of the persistent and intrusive negative thoughts scale (PINTS). *Psychol Assess*. 2019;31:1329.
49. Strzelecki A. To use or not to use ChatGPT in higher education? A study of students' acceptance and use of technology. *Interact Learn Environ*. 2023;1–14.
50. Venkatesh V, Thong JYL, Xu X. Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS Q*. 2012;157–78.
51. Podsakoff PM, MacKenzie SB, Lee J-Y, Podsakoff NP. Common method biases in behavioral research: a critical review of the literature and recommended remedies. *J Appl Psychol*. 2003;88:879.
52. Hair JF, Black WC, Babin BJ, Anderson RE, Tatham RL. Multivariate data analysis. Volume 6. Pearson Prentice Hall Upper Saddle River; 2006.
53. Hair JF Jr, Matthews LM, Matthews RL, Sarstedt M. PLS-SEM or CB-SEM: updated guidelines on which method to use. *Int J Multivar Data Anal*. 2017;1:107–23.
54. Ringle CM, Sarstedt M. Gain more insight from your PLS-SEM results: the importance-performance map analysis. *Ind Manag Data Syst*. 2016;116:1865–86.
55. Fornell C, Larcker DF. Structural equation models with unobservable variables and measurement error: Algebra and statistics. 1981.
56. Foltýnek T, Meuschke N, Gipp B. Academic plagiarism detection: a systematic literature review. *ACM Comput Surv*. 2019;52:1–42.
57. Mumford MD, Waples EP, Antes AL, Brown RP, Connelly S, Murphy ST, et al. Creativity and ethics: the relationship of creative and ethical problem-solving. *Creat Res J*. 2010;22:74–89.

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